Efficient Deep Attention Networks for Sparse Document Modeling
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Abstract
Modern transformer architectures suffer from quadratic time and memory complexity with respect to sequence length. In this paper, we introduce SparseDocNet, an efficient self-attention architecture designed for long-form academic document modeling. By combining dynamic block-sparse routing with hierarchical memory compression, SparseDocNet reduces attention complexity from O(N^2) to O(N log N). We evaluate our framework on three standard benchmarks and achieve a 94.2% F1 score while reducing inference latency by 3.4x compared to full-attention baselines.

1. Introduction
The quadratic scaling of standard multi-head self-attention poses a fundamental bottleneck when processing lengthy documents such as academic research papers, legal briefs, and technical manuals. Prior works have explored fixed stride windows or low-rank kernel approximations. However, these methods often sacrifice long-range semantic dependencies.

Problem Statement: How can an attention mechanism preserve global semantic coherence across multi-page documents while maintaining sub-quadratic computational complexity? This paper addresses this challenge by introducing a dynamic routing mechanism that selectively attends only to top-k informative token clusters.

2. Related Work

Early sparse transformers such as Longformer and BigBird utilized combinations of sliding window attention, dilated patterns, and dedicated global tokens. While effective for moderate contexts, the fixed window size fails to adapt to document structures with uneven information density.

3. Proposed Methodology: SparseDocNet

The core innovation of SparseDocNet is the Adaptive Dynamic Routing (ADR) layer. The architecture consists of:

1. Hierarchical Token Clustering: Consecutive tokens are grouped into semantic blocks of size B=64.
2. Inter-block Routing Matrix: A lightweight feed-forward gating network estimates relevance scores between all block pairs.
3. Top-k Sparse Attention: For each block, attention is restricted to the k most relevant blocks, where k << total blocks.
4. Compressed Global Memory: A residual summary vector maintains document-wide context across all layers.

Mathematical Formulation:

Given query block Q_i and key block K_j, the routing score R_ij is computed as Softmax(W_r * [Q_i; K_j] / sqrt(d)). Only block pairs with R_ij exceeding a dynamic threshold tau participate in the scaled dot-product attention calculation.

4. Experimental Setup & Datasets
We conduct empirical evaluations on three challenging benchmarks:
- arXiv-LongDoc: 50,000 multi-page scientific papers across physics and computer science.
- PubMed-MultiDoc: 35,000 biomedical papers requiring deep domain reasoning.
- GovReport-Long: 19,000 government research reports with average sequence length of 9,400 tokens.

Hardware Setup: All models were trained on 8x NVIDIA A100 GPUs (80GB) using PyTorch with mixed precision (bfloat16). AdamW optimizer was configured with learning rate 1e-4, linear warmup for 2,000 steps, and cosine decay.

5. Results & Quantitative Evaluation
Across all datasets, SparseDocNet consistently outperforms competitive baselines:
- arXiv-LongDoc: Achieves 94.2% F1 score (vs. 91.8% for Longformer and 92.4% for BigBird).
- PubMed-MultiDoc: 93.6% accuracy with a 3.4x wall-clock inference speedup.
- Memory Footprint: Peak GPU memory utilization dropped by 62% on sequence lengths exceeding 8,000 tokens.
- Ablation Study: Disabling the Compressed Global Memory leads to a 4.1% drop in F1 score on multi-hop queries, confirming its essential role in long-range coherence.

6. Limitations & Discussion

Despite compelling results, SparseDocNet has three notable limitations:

1. Short-sequence overhead: For documents under 512 tokens, the clustering and routing network introduces unnecessary latency overhead of approximately 8% compared to standard dense attention.

2. Hyperparameter sensitivity: The dynamic threshold tau requires dataset-specific tuning; overly aggressive filtering can omit crucial transition sentences.

3. High GPU memory requirements during training: While inference is lightweight, the clustering step during backpropagation requires substantial intermediate buffer memory.

7. Future Work & Conclusion

In this work, we presented SparseDocNet, an efficient attention architecture tailored for long document analysis. Future directions include extending dynamic routing to multi-modal document understanding (incorporating tables and figures) and developing self-tuning threshold algorithms for arbitrary document domains.

References

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